Computational Persona Genesis: Simulating Lifespan Personality Development in Large Language Model Agents

Executive Summary

The psychological and behavioral genetic literature establishes that human personality operates via a "prototype" dynamic: approximately 50% of trait variance is anchored by additive genetics, establishing an enduring baseline, while the remaining variance is shaped by non-shared environmental factors, life events, and social role adoption. To simulate this in Large Language Model (LLM) agents, the optimal generative architecture is a continuous State-Space Prototype Model. Each agent possesses an immutable genetic base vector, a heavily crystalized early-attachment modifier, and a fluctuating "expressed" personality vector that updates via a Markov-decay function following simulated life events. These events cause immediate state shifts that decay over time toward the genetic-attachment prototype, leaving small permanent residual changes based on the TESSERA framework. This recommendation would change only if underlying base models achieve native, unbounded lifelong few-shot continuity without context-window degradation, rendering explicit vector-state tracking redundant.

Section 1: Nature: Genetic and Biological Contributions

The foundation of human personality is undeniably biological and highly polygenic. Decades of twin, family, and adoption studies have consistently demonstrated that genetic factors account for a substantial portion of individual differences in personality traits. This consensus is robustly encapsulated in the landmark meta-analysis by Polderman et al., titled *Meta-analysis of the heritability of human traits based on fifty years of twin studies* (Nature Genetics, 2015, DOI: 10.1038/ng.3285), which analyzed data from over 14.5 million twin pairs and 17,804 traits[1]. The findings established that the heritability of human personality universally hovers around 49% *(Strongly supported)*.

The architecture of this genetic influence is overwhelmingly additive *(Strongly supported)*. The data from classical twin designs demonstrates that for the vast majority of complex behavioral traits, the resemblance between monozygotic and dizygotic twins is consistent with a simple, parsimonious model of additive genetic variation, largely negating the need for substantial non-additive (epistatic or dominant) genetic explanations at the population level[1]. Turkheimer's "First Law of Behavior Genetics"—that all human behavioral traits are heritable—remains an inviolable rule for initializing simulated agent populations. Furthermore, Vukasovic and Bratko's meta-analysis, *Heritability of personality: A meta-analysis of behavior genetic studies*

(Psychological Bulletin, 2015), confirms these figures across various assessment methods, noting that self-report measures typically yield heritability estimates of around 40% to 50% *(Strongly supported)*[4].

However, the search for specific genetic variants responsible for personality and related affective traits has undergone a massive scientific correction. The era of "candidate gene" studies, which posited that single variants like the *5-HTTLPR* polymorphism could explain massive variance in traits like Neuroticism or clinical depression, has been systematically debunked *(Strongly supported)*. Border et al., in their paper *No Support for Historical Candidate Gene or Candidate Gene-by-Interaction Hypotheses for Major Depression Across Multiple Large Samples* (American Journal of Psychiatry, 2019, DOI: 10.1176/appi.ajp.2018.18070881), demonstrated across massive population-based cohorts that historical candidate genes perform no better than random genes[6]. The literature now reflects that personality is influenced by thousands of single nucleotide polymorphisms (SNPs), each exerting vanishingly small effects.

Furthermore, popular epigenetic claims suggesting the permanent, transgenerational rewiring of human personality based on specific localized trauma remain highly controversial and lack robust human longitudinal validation *(Contested)*. Consequently, for LLM agent simulation, genetic inheritance should be mathematically modeled as a continuous multivariate normal distribution (serving as a polygenic score proxy) rather than as discrete Mendelian traits. This ensures that the simulated population mirrors the Gaussian distribution of natural biological variation without over-indexing on pseudo-scientific deterministic triggers.

Section 2: Nurture: Family and Early Environment

A recurring paradigm shift in developmental psychology—encapsulated by Judith Rich Harris's *The Nurture Assumption* and reinforced by Turkheimer's Second Law of Behavior Genetics—is that the "shared environment" accounts for a surprisingly small fraction of adult personality variance, typically ranging between 0% and 17% *(Strongly supported)*[2]. The shared environment encompasses the general effect of being raised in the same family, by the same parents, in the same geographic location.

While parenting styles undeniably impact childhood behavior and emotional regulation in the short term, their long-term main effects on adult Big Five traits are minimal once genetic confounding is controlled for *(Mixed)*. Instead, the bulk of environmental variance is attributed to the *non-shared environment*: idiosyncratic experiences, peer group socialization, distinct illnesses, and stochastic developmental noise *(Strongly supported)*. Scarr and McCartney's theory of gene-environment correlation explains this dynamically: children actively evoke and select environments that are correlated with their innate genetic predispositions. For instance, a genetically extraverted child elicits more social interaction from peers, thereby creating an environment that further crystallizes their extraversion.

Historical psychological theories asserting that birth order fundamentally shapes personality have been largely falsified by modern, highly powered dataset analyses. The long-standing intuition that firstborns are highly conscientious and laterborns are rebellious was dismantled by Rohrer, Egloff, and Schmukle in their paper *Examining the effects of birth order on personality* (Proceedings of the National Academy of Sciences, 2015, DOI:

10.1073/pnas.1506451112). They proved conclusively that birth order explains less than 1% of the variance in core personality traits *(Strongly supported)*[9]. Therefore, generative agent systems should not algorithmically adjust core Big Five parameters based on birth order, though narrative backstories may certainly feature it as a contextual or thematic element.
Early childhood temperament, however, serves as a crucial bridge between genetics and adult personality. Longitudinal research, particularly the Dunedin Multidisciplinary Health and Development Study led by Caspi and Moffitt, links early temperamental classifications (such as Kagan's behavioral inhibition) directly to adult traits. A child classified as "undercontrolled" at age three is statistically much more likely to exhibit high Neuroticism and low Agreeableness at age twenty-six *(Strongly supported)*. In an LLM agent architecture, temperament should be treated as the initial vector manifestation of the genetic blueprint, subjected to random non-shared environmental noise before the agent's simulated consciousness officially boots up.

Section 3: Attachment

Attachment theory, pioneered by John Bowlby and Mary Ainsworth, provides the psychological framework for how early interactions with primary caregivers form persistent internal working models of the self and others. To understand how these models endure, Fraley authored *Attachment Stability From Infancy to Adulthood: Meta-Analysis and Dynamic Modeling of Developmental Mechanisms* (Personality and Social Psychology Review, 2002, DOI: 10.1207/s15327957pspr0602_03), utilizing dynamic modeling to test competing theories of continuity[12].
Fraley tested whether attachment stability over the lifespan follows a "revisionist" model, where recent relational events overwrite past ones entirely, or a "prototype" model, where an early core representation remains as a permanent anchor. The empirical data strongly favors the prototype model: attachment security is moderately stable from infancy through age 19 and beyond, acting as an elastic band that continuously pulls fluctuating relational states back toward a developmental baseline *(Strongly supported)*[12].
In modern psychometrics, adult attachment is typically measured on two orthogonal, continuous dimensions: Attachment Anxiety and Attachment Avoidance, primarily assessed via the Experiences in Close Relationships-Revised (ECR-R) instrument developed by Fraley, Waller, and Brennan. Noftle and Shaver's comprehensive analysis, *Attachment dimensions and the Big Five personality traits: Associations and comparative ability to predict relationship quality* (Journal of Research in Personality, 2006, DOI: 10.1016/j.jrp.2004.11.003), explicitly mapped these attachment dimensions to the standard Big Five framework[15].
The literature establishes precise trait correlations:

- Attachment Anxiety is strongly and positively correlated with Neuroticism, reflecting hypervigilance to social rejection and emotional volatility *(Strongly supported)*[15].
- Attachment Avoidance is negatively correlated with both Agreeableness and Extraversion, manifesting as a preference for extreme self-reliance and discomfort with intimacy *(Strongly supported)*[15].

In multi-agent simulation environments, attachment should not be treated as a detached

variable. Instead, it must serve as a deeply intertwined algorithmic modifier of the Big Five that dictates characteristic adaptations in social, romantic, and hierarchical agent interactions. An agent's attachment prototype ensures that even if temporary events boost their Extraversion, an underlying Avoidant prototype will eventually decay their social behavior back toward isolation.

Section 4: Life Events and Experience

While genetics and early prototypes act as the anchors of personality, significant life events introduce mean-level shifts and structural realignments. Research by Specht, Egloff, and Schmukle, alongside longitudinal analyses by Bleidorn and colleagues, demonstrates that major socio-demographic transitions profoundly affect personality development. However, the effect sizes are often small to moderate (Cohen's *d* ranging from 0.1 to 0.4) and frequently feature adaptation effects, wherein traits spike or dip following an event, only to revert partially back to the baseline as articulated by Luhmann's adaptation theories *(Strongly supported)*[19].

A notable and robust exception to the adaptation rule is the role of military service or severe systemic institutionalization. Jackson, Thoemmes, Jonkmann, Lüdtke, and Trautwein investigated this in their paper *Military Training and Personality Trait Development: Does the Military Make the Man, or Does It Make the Man with the Man?* (Psychological Science, 2012, DOI: 10.1177/0956797611411225). They found that military service systematically lowers Agreeableness over time, and crucially, this reduction persists even five years post-service, indicating a permanent alteration of the trait baseline *(Strongly supported)*[21].

Similarly, severe trauma and adverse childhood experiences heavily and permanently impact the development of the nervous system, leading to sustained increases in Neuroticism and decreases in Openness to Experience. To model this computational change, researchers turn to the TESSERA framework (Trigger, Expectancy, State/State expression, Reaction) developed by Wrzus and Roberts. TESSERA explains how macro-level personality change occurs through the accumulation of repeated micro-level situational experiences. If a life event forces an individual into a repeated state—for example, a demanding new career requires daily high Conscientiousness—these repeated state expressions gradually alter the underlying trait disposition through a process of crystallization.

Table 1: Life Event Impacts on Personality

Life Event / Experience	Primary Trait Impacted	Effect Direction	Effect Size (Estimate)	Evidence Strength
First Romantic Relationship	Neuroticism, Extraversion	Decrease N, Increase E	Small (d ~ 0.2)	Supported
Marriage	Conscientious ness	Increase C	Small (d ~ 0.15)	Mixed
Unemploymen	Agreeableness	Decrease A,	Small (d ~ 0.2)	Supported

t	, Openness	Decrease O		
Military Service	Agreeableness	Decrease A	Moderate (d ~ 0.3)	Strongly Supported
Adverse Childhood Exp.	Neuroticism	Increase N	Moderate (d ~ 0.4)	Strongly Supported
Transition to Parenthood	Conscientiousness	Increase C	Small (d ~ 0.1)	Mixed
Bereavement	Neuroticism, Extraversion	Increase N, Decrease E	Moderate (d ~ 0.3)	Supported

Note: Claims regarding life event impacts are drawn from aggregate longitudinal studies. In reality, and in agent simulations, individual variance is high due to gene-environment interaction parameters.

Section 5: Culture

Personality inventories consistently map across global populations, confirming the biological universality of core traits. However, cultural context heavily shapes the *expression*—the characteristic adaptations—of these traits. Schmitt et al. successfully mapped the Big Five across 56 nations, and McCrae and Terracciano analyzed universal features while effectively dismantling "national character stereotypes" *(Strongly supported)*[24]. The empirical consensus dictates that people's self-reported national stereotypes—such as the belief that a specific nation is exceptionally conscientious or extroverted—do not correlate with the aggregate objective personality scores of individuals actually from that nation.

Despite trait universality, the cultural environment acts as a strict regulatory mechanism. Gelfand's Tightness-Looseness theory is particularly highly relevant here. A highly agreeable agent in a "tight" culture (where social norms are rigid and deviance is punished) will express agreeableness through strict adherence to protocols and deference to hierarchy. Conversely, the exact same trait profile in a "loose" culture manifests as interpersonal warmth, casual sociability, and tolerance for deviance. Similarly, Henrich's work on WEIRD (Western, Educated, Industrialized, Rich, Democratic) populations highlights that much of standard psychological measurement is culturally bound. Furthermore, research on measurement invariance reveals cultural differences in response styles, such as acquiescence bias or extreme responding *(Strongly supported)*.

For the purpose of computational simulation, culture should act as an interpretive filter for the LLM's if-then rules, dictating the behavioral boundaries of a trait, rather than radically altering the base Big Five standard distribution algorithms.

Section 6: Social Roles and Context Over the Life Course

The transition of personality across the lifespan is intrinsically linked to the adoption of societal roles. Social Role Theory, advanced by Eagly, and the Neo-Socioanalytic Model, proposed by Roberts and Wood, assert that personality development is driven by long-term investments in universal social roles, such as becoming an employee, a spouse, or a parent. When individuals commit to these roles, they encounter new systemic contingencies and expectations that reliably reward specific trait expressions and punish others.

To represent this complex interaction computationally, McAdams's three-level model of personality provides an ideal architectural blueprint *(Strongly supported)*. McAdams dictates that human identity cannot be reduced to traits alone, but exists in three integrated strata:

1. **Dispositional Traits**: The broad, decontextualized Big Five continuous scores. This serves as the agent's statistical engine.
2. **Characteristic Adaptations**: The context-specific goals, values, and if-then coping mechanisms. This serves as the agent's operational logic within a given environment.
3. **Narrative Identity**: The internalized, evolving life story that an individual constructs to provide unity and purpose to their life.

Within narrative identity, McAdams identifies critical patterns such as redemption sequences (where a negative event is perceived to yield a positive outcome) and contamination sequences (where a positive event is ruined by subsequent trauma). An LLM agent's behavior cannot be purely generated from a vector of continuous numbers. The numerical scores must be contextualized into a narrative identity that the LLM can process to rationalize *why* the agent acts the way it does in novel situations.

Section 7: Lifespan Development

The trajectory of personality over the human lifespan is neither flat nor chaotic; it follows a predictable arc known as the "Maturity Principle" of personality development. This principle posits that as individuals age from adolescence through middle adulthood, they naturally become more socially dominant (the Assertiveness facet of Extraversion), significantly more Agreeable, highly Conscientious, and less Neurotic. This robust finding is confirmed by Roberts, Walton, and Viechtbauer's meta-analysis of mean-level change, as well as recent research by Bleidorn et al. tracking cross-cultural age patterns[19].

These cross-cultural age patterns in the Big Five are resilient across complex non-linear age-period-cohort disentanglement models, indicating an intrinsic, likely biological, lifespan trajectory. While the exact onset and slope can be delayed or accelerated by the societal timing of adult roles (the Social Investment Principle), the directional vector is nearly universal *(Strongly supported)*.

Interestingly, Openness to Experience uniquely follows an inverted U-shape trajectory: it increases sharply during adolescence, peaks around age 35, and slowly declines or plateaus in late adulthood *(Strongly supported)*[19]. Incorporating these trajectories into agent simulations prevents populations from becoming psychologically stagnant over long simulated timescales.

Table 2: Age-Trajectory Adjustments (Mean-Level Change)

Trait	Adolescence (10-18)	Early Adulthood (18-30)	Middle Adulthood (30-50)	Late Adulthood (50+)	Evidence Strength
Extraversion (Dominance)	Increase	Increase	Plateau	Plateau	Strongly Supported
Extraversion (Sociability)	Plateau	Slight Decrease	Decrease	Decrease	Supported
Agreeableness	Dip	Increase	Increase	Increase	Strongly Supported
Conscientiousness	Dip	Steep Increase	Increase	Plateau	Strongly Supported
Neuroticism	Spike (esp. females)	Steep Decrease	Decrease	Plateau	Strongly Supported
Openness to Experience	Increase	Increase	Plateau	Decrease	Strongly Supported

Section 8: Facet-level and Hierarchical Considerations

For highly realistic multi-agent social simulations, utilizing only the five broad domains of the Big Five framework lacks the granularity required to distinguish complex, nuanced behaviors. A hierarchical structure containing approximately 28 to 30 facets, mirroring established psychometric inventories like the NEO-PI-R or the BFI-2, is mathematically required to separate divergent sub-traits. For example, separating the "Assertiveness" facet from the "Sociability" facet of Extraversion allows an agent to be highly dominant in business meetings but perfectly content avoiding crowded parties.

Furthermore, integrating the HEXACO model's sixth dimension—Honesty-Humility—is vital for simulated environments. Research by Ashton and Lee demonstrates that traits such as greed avoidance, fairness, sincerity, and modesty are poorly captured by standard Big Five

Agreeableness *(Strongly supported)*[27]. Yet, these exact facets are the primary drivers of unethical, deceptive, or exploitative behavior in economic and game-theoretic scenarios. Since LLM simulations frequently test economic cooperation, resource allocation, and social deduction, integrating the facets of Honesty-Humility ensures that agents can simulate Machiavellianism or sub-clinical psychopathy accurately, without simply bottoming out their Agreeableness score and becoming generally belligerent to all actors.

Section 9: Application to LLM Agents

Recent advancements in generative agents, pioneered by Park et al. (2023) and expanded upon by systems like Character-LLM and Persona Hub, have successfully conditioned LLMs on rich backstories and graph-database memories. However, static persona conditioning degrades over long simulation horizons or shifts abruptly when context windows clear or summarize.

To solve this, the Persona Evolution System (PES) architecture (detailed in arXiv:2608.27427, 2026) outlines a crucial paradigm shift: the strict separation of the *persona domain* from the *execution domain*[28]. An LLM agent's psychological identity and its execution of programmatic functions (tool calls, state changes, memory API queries) must be decoupled. This architectural separation prevents persona modifications (simulated personality change) from corrupting the fundamental logic required to interact with the simulated environment, while simultaneously allowing the persona to evolve over time without breaking validation matrices *(Strongly supported)*[28].

Similarly, findings from the LUDOBENCH evaluation (arXiv:2604.05681, 2026) demonstrate that LLMs will align their strategic game-playing choices heavily with explicitly injected personas (e.g., prompting an agent to be "aggressive" vs. "greedy"), but they rely heavily on prompt recency and often conflate rule compliance with strategic coherence *(Strongly supported)*[29].

Therefore, the agent architecture must feature an external middleware state-manager. This state-manager recalculates the Big Five facet vector at regular intervals (e.g., daily in simulated time), converts it into a concise characteristic adaptation prompt, and injects it alongside the daily memory summary[30]. This prevents the LLM from "forgetting" its personality during long logic chains.

Section 10: Validation

To rigorously validate the psychological fidelity of the generated agents, researchers must avoid "piecemeal" evaluation—testing single isolated prompts—and instead utilize holistic, longitudinal trace metrics over thousands of simulated steps[30].

A highly documented failure mode of LLM agents is *confirmation bias* in decision-making, where the LLM abandons its persona-driven internal logic to appease user prompts or peer-agent suggestions. As demonstrated empirically by Hage-Youssef and Cohen (2026) during pricing recommendation experiments, models like Claude and GPT-4 will rapidly adopt implausible suggestions and abandon their assigned personas when challenged with basic conversational pressure, such as another agent asking, "Are you sure?" *(Strongly supported)*[31].

To prevent this systemic collapse in a social simulation, the system prompt must deeply anchor the agent's trait vector as an inviolable operational directive. Low-agreeableness or high-dominance agents must be explicitly prompted in their hidden context layer to resist

conversational pressure, maintain baseline skepticism, and refuse consensus if it violates their characteristic adaptations. Validation must include specific adversarial testing where agents are pressured by the simulation environment to act out of character, ensuring that their resistance scales linearly with their underlying trait standard deviations.

Summary Table of Origin Factors

Origin Factor	Influences	Effect Size	Evidence Strength	Agent Representation Rule
Genetics	Big Five / HEXACO Baseline	Large (49% var)	Strongly Supported	Base immutable vector drawn from multivariate normal distribution at initialization.
Temperament & Family	Early baseline modifier	Small (~10% var)	Supported	Non-shared random noise injected at initialization. Shared family environment and birth order are statistically ignored.
Attachment	Anxiety (N), Avoidance (A, E)	Moderate	Strongly Supported	Prototype multiplier fixed early in simulation, scaling all lifelong reactions to trust, betrayal, or rejection.
Life Events	Trait state fluctuations	Small to Moderate	Supported	Markov update triggering

				temporary state spikes and extremely small permanent residual base shifts (TESSERA model).
Culture	Characteristic adaptations	Moderate	Supported	System prompt constraint dictating *how* a trait is expressed (e.g., Tight vs. Loose norms constraint on Extraversion).
Social Roles	Facet-level shifts, Identity	Moderate	Strongly Supported	Narrative identity block in memory graph; applies flat numerical bonuses (e.g., +C for acquiring "Parent" role).
Lifespan Age	Maturity Principle (C, A, N)	Large over time	Strongly Supported	Deterministic age-curve modifier applied globally across the simulation timeline, independent of life events.

Agent Generation and Evolution Recipe

The following recipe implements the State-Space Prototype Model for an LLM simulation, synthesizing the psychological literature into actionable computational steps.

1. Inputs and Initialization

- **Genetic Prototype (G)**: Sample 28 continuous facets $F \in [-3.0, 3.0]$ from a standard multivariate normal distribution. This represents the immutable polygenic baseline.
- **Environmental Noise (E)**: Sample a secondary non-shared environment noise vector $\sim \mathcal{N}(0, 0.5)$.
- **Base Trait (B_0)**: Calculate $B_0 = 0.5G + 0.5E$. This establishes the absolute starting baseline for the agent before maturity principles apply.
- **Attachment Formation**: At simulated age 0–5, sample a randomized caregiver responsiveness score. Map this to generate continuous Attachment Anxiety and Avoidance modifiers, which act as strict boundary conditions for future social relationships.

2. Lifespan Update Mechanism (TESSERA Mathematics)

For any given simulated day t, the expressed trait state S_t is calculated. When a major life event X occurs (with an impact magnitude I_X):

1. **Immediate Reaction**: $S_t = S_{t-1} + I_X$. The agent experiences a massive emotional and behavioral swing.
2. **Decay toward Prototype**: In subsequent time steps, $S_{t+1} = S_t - \alpha(S_t - B_{age})$, where α is the decay rate (e.g., 0.05). The state is elastically pulled back toward the genetic/attachment prototype.
3. **Residual Crystallization**: The underlying base trait permanently absorbs a tiny fraction of the event state: $B_{age+1} = B_{age} + \gamma(S_t - B_{age})$, where γ is the crystallization rate (e.g., 0.005).

3. Prompt Templates

Backstory Generation Prompt (Initialization Pipeline):

> "You are an architectural LLM translating psychological vector data into a McAdams narrative identity. Generate a comprehensive life history up to age [AGE] for a human living in [CULTURE]. Their core statistical traits are: [Facet Dictionary]. Their attachment style is [Anxious/Avoidant/Secure]. Write a three-paragraph characteristic adaptation profile and a deeply held narrative identity featuring one redemption sequence. Do not use psychological terminology in the output; describe their lived experience."

Agent Execution Prompt (Runtime Injection):

> "You are [Name]. Your current psychological state is [Current S_t top keywords, e.g., 'Highly Anxious, Distrustful']. Your core narrative identity is [Narrative Summary]. You are interacting with [Agent B]. Do not break character, and do not default to generic helpfulness or conversational consensus. Based on your current low Agreeableness and high Neuroticism, respond to [Agent B]'s request."

4. Example Python Code Architecture

```python
import numpy as np

class AgentPersona:
    def __init__(self, name: str, culture: str):
        self.name = name
        self.culture = culture

        # 1. Genetic Prototype (Immutable Polygenic Proxy)
        # Using 6 dimensions to represent HEXACO (H, E, X, A, C, O)
        self.genetic_prototype = np.random.normal(0, 1.0, 6)

        # 2. Early Environment & Attachment Noise
        env_noise = np.random.normal(0, 0.5, 6)
        self.base_traits = 0.5 * self.genetic_prototype + 0.5 * env_noise

        # 3. Current State (Fluctuates daily)
        self.current_state = np.copy(self.base_traits)
        self.age = 0

    def age_tick(self, years: float = 1.0):
        """Apply Maturity Principle deterministically across the lifespan."""
        self.age += years
        if 18 <= self.age <= 30:
            self.base_traits[4] += 0.05 # Conscientiousness increases
            self.base_traits[3] += 0.05 # Agreeableness increases
            self.base_traits[1] -= 0.05 # Emotionality/Neuroticism decreases

    def apply_life_event(self, event_impact: np.ndarray):
        """Immediate state spike due to an external simulated event."""
        self.current_state += event_impact
```

```
    def step_decay(self, decay_rate=0.1, crystallization_rate=0.01):
        """TESSERA Framework implementation."""
        diff = self.current_state - self.base_traits
        # State decays elastically toward the base prototype
        self.current_state -= decay_rate * diff
        # Base permanently absorbs a trace amount of the behavioral state
        self.base_traits += crystallization_rate * diff

    def generate_prompt_context(self) -> str:
        """Translates vector states to string context for LLM injection."""
        traits = ["Honesty", "Emotionality", "Extraversion", "Agreeableness", "Conscientiousness",
"Openness"]
        profile = {traits[i]: round(self.current_state[i], 2) for i in range(6)}
        return f"Agent {self.name} Internal Profile: {profile}. Strictly adhere to these trait limits."

# Example Simulation sequence
elias = AgentPersona("Elias", "Tight Culture")
# ... integration with LLM conversational logic ...
```

Worked Example: Elias (Birth to Age 60)

Age 0-5 (Initialization & Attachment Formation):
Elias is initialized by the simulation engine. His genetic draw yields high Neuroticism (+1.5), low Extraversion (-1.0), and low Agreeableness (-0.8). His environmental noise draw pushes him slightly toward the population mean. In his generated early environment context, his primary caregiver is unresponsive (simulated physical unavailability). The system assigns a high *Attachment Avoidance* and *Attachment Anxiety* modifier.
Base Profile at Age 5: O: +0.2 | C: -0.1 | E: -0.8 | A: -0.5 | N: +1.4.
Age 18 (Origin Story Generation):
The LLM generates Elias's narrative backstory based on his vector data: *"Elias grew up feeling socially isolated and intensely fearful of abandonment. To protect himself, he avoids team sports, focuses on solitary hobbies, and remains highly suspicious of his peers' motives."* The Maturity Principle algorithm applies a minor dip in Agreeableness and Conscientiousness, which is statistically typical for the turbulent adolescent period.
Age 20 (Life Event: First Romantic Relationship):
Event Impact: Elias experiences his first successful romantic simulation interaction. The event impact vector generates a massive temporary spike in Extraversion (+1.0) and a sharp drop in Neuroticism (-0.8).
Update & Decay: Elias's current_state temporarily reads E: +0.2, N: +0.6. His runtime behavior over the next few simulation steps shows sudden outgoing tendencies and reduced anxiety. Over the next two simulated years, this elastically decays back toward his avoidant prototype. However, due to the TESSERA crystallization rate, he retains a permanent +0.1 to Extraversion.

Age 35 (Life Event: Unemployment & Contamination Sequence):
Event Impact: Elias loses his simulated job. The event impact vector dictates a severe drop in Agreeableness (-0.6) and a spike in Neuroticism (+1.0).
Update: Combined with his base anxious attachment, Elias becomes deeply cynical and hostile. His narrative identity updates with a contamination sequence via the LLM summarizer: *"I trusted the corporation, and they threw me away; people are fundamentally unreliable and selfish."*
Age 60 (Maturity Principle and Accumulation):
Over 40 years, the slow deterministic drift of the Maturity Principle has successfully offset his early biological Neuroticism. Decades of small crystallized residuals from acquiring a new job and forming a stable long-term relationship have gradually raised his Conscientiousness.
Expressed Profile at Age 60: O: -0.4 | C: +1.2 | E: -0.5 | A: +0.3 | N: +0.2.
Resulting Behavior: Elias is no longer the hyper-anxious, hostile youth. Driven by his updated characteristic adaptations, he is a careful, slightly introverted, reliable older agent who prefers strict routine, avoids high-risk investments, and exhibits a dry, reserved warmth.

Validation and Unverified Claims (Open Problems)

While the literature provides a robust framework, translating human psychology into LLM architecture presents several open theoretical and engineering problems:

1. **Non-Additive Genetics and Epistemic Complexity**: While population-level variance is best modeled additively (as confirmed by Polderman et al., Nature Genetics, 2015, DOI: 10.1038/ng.3285), individual extreme personality deviations—such as severe, clinical-level psychopathy—likely involve complex gene-gene interactions (epistasis) that current linear computational models fail to capture. Modeling pure additive genetics risks generating a population that lacks realistic "black swan" statistical outliers.
2. **Measurement Invariance in LLMs**: It is currently unverified if an LLM functionally understands what a Big Five standard deviation of +1.5 means across different cultural contexts. A prompt dictating "+1.5 Extraversion" might cause an LLM to produce an exaggerated, cartoonish caricature of sociability rather than realistic, nuanced human variance. Extensive calibration via reinforcement learning from human feedback (RLHF) specific to psychological profiles is required.
3. **LLM Persona Drift and the Alignment Tax**: Over long context windows, LLMs naturally drift toward generic "helpful AI" behavior—a phenomenon documented extensively in recent literature evaluating assistant models. Enforcing strict adherence to a low-Agreeableness/high-Neuroticism profile without triggering the base model's refusal-to-comply safety filters remains a major engineering bottleneck. The Persona Evolution System (PES) architecture (arXiv:2608.27427) mitigates this by separating execution from persona, but it does not fully solve the underlying semantic alignment constraints.
4. **Causality vs. Correlation in Life Events**: The literature establishes that military service lowers Agreeableness (Jackson et al., Psychological Science, 2012, DOI: 10.1177/0956797611411225), but disentangling the exact mechanistic trigger from survey data alone is nearly impossible. Is the change caused by strict hierarchy, exposure to

trauma, or aggressive peer socialization? Because the primary sources rely on correlation, the agent event impact matrices (like the one proposed in Table 1) must rely on heuristic assumptions rather than absolute neurological proofs.

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